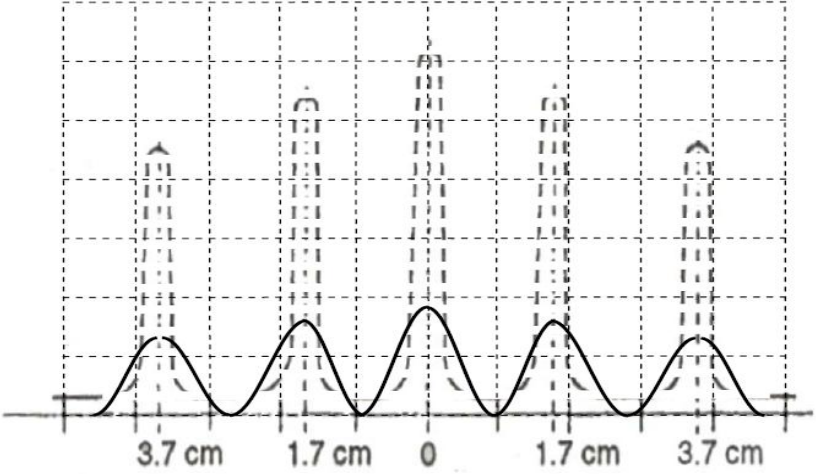


8ai	$v = f\lambda$ $0.20 = \left(\frac{1}{0.60}\right)\lambda$ $\lambda = 0.12 \text{ m}$	A1	
8aii	Phase difference = $(0.20/0.60) \times 2\pi = 2.09 \text{ rad}$ Or Phase difference = $(0.40/0.60) \times 2\pi = 4.19 \text{ rad}$	A1	
8aiii	Intensity $I \propto A^2$ $I_Q / I_P = A_Q^2 / A_P^2$ $= 2.0^2 / 3.0^2$ $= 0.44$	C1 A1	
8aiv	Displacement = $1.00 - 3.00 = -2.00 \text{ mm}$	A1	
8bi	It means the two waves have <u>constant phase difference</u> .	B1	
8bii	P is the second minima from the central bright fringe, Path difference = $1.5\lambda = 1.5 \times 610$ $= 920 \text{ nm}$	A1	
8biii	$\lambda = ax / D$ $x = (0.022 / 4) = 0.0055 \text{ m}$ $a = (610 \times 10^{-9})(2.7) / (0.0055)$ $= 3.0 \times 10^{-4} \text{ m}$	C1 A1	

8biv	When visible light of higher frequency is used, the <u>wavelength becomes shorter</u> . Since <u>fringe separation is directly proportional to wavelength</u> for the same slit separation and distance D, the <u>fringe separation decreases</u> .	B1 B1	
8bv	No change to fringe separation / fringe width / number of fringes Bright fringes are brighter, Dark fringes unchanged.	B1 B1	
8ci	When the turntable rotates, <u>the second order of the diffraction pattern will be registered first by the detector, forming peak A, followed by the first order as peak B and then the central maximum as peak C</u> . As the table continue to rotate, the other first order on the other side of the central maximum will be registered as peak D, followed by the second order as peak E. As the turntable rotates at constant angular speed, the other half of the peaks (D and E) <u>appear at same separation away from the central maximum</u> as peaks B and A respectively.	B1 B1	
8cii1	Time to shift from zeroth order to peak D through 1.7 cm on the screen is $1.7 \times 0.10 = 0.17$ s $\theta = \omega t = (2\pi/3.0) (0.17)$ $\theta = 0.356$ rad	M1 A0	
8cii2	$d \sin \theta_1 = n \lambda$ $(\frac{10^{-3}}{550}) \sin(0.356) = (1) (\lambda)$ $\lambda = 6.34 \times 10^{-7}$ m	C1 A1	
8ciii	 Either or - Since the slit separation remain the same, the <u>fringe separation remains the same</u>. - Using double slit equation, fringe separation is approximated to be equal. Thus, 2nd order peak at 3.4 cm is allowed.	B1 B1	

	However, the <u>peaks are less intense</u> (less contrast between the bright and dark fringes) and <u>less sharp</u> (broader and less defined)		
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